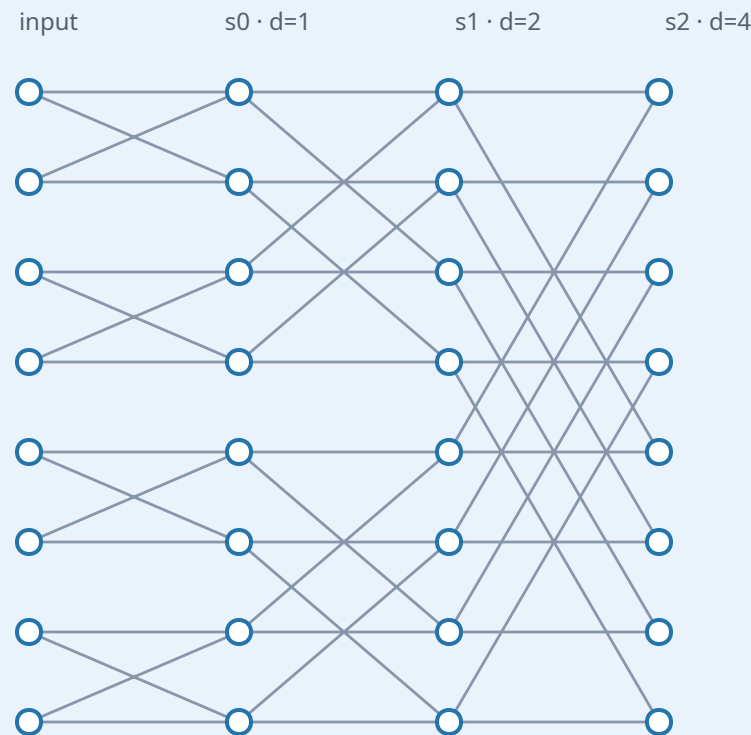


A concrete N=8 FFT: what Ud, Td, Us and Ts change

The factors describe resource replication and reuse. They do not fix the tile size, radix or physical core.

Logical dependency graph

stage-space points right; data groups run vertically



same stage: independent butterfly groups
next stage: prescribed partner dependency

Four mappings of the same graph

Ud = 2 · data space

replicate independent workers

AIV 0
batch b0

AIV 1
batch b1

no dependency between transforms

Td = 2 · data time

reuse one worker over data

AIV 0
b0



same
b1

amortizes setup; adds steady work

Us = 2 · stage space

replicate stage service roles

role 0
stage s0



role 1
stage s1

requires transport + readiness

cross-AIV role pipeline is not production

Ts = 3 · stage time

reuse service while state resides

s0



s1



s2

current short path keeps stages in UB

state capacity bounds the factor